

$\sum_{i=1}^n (x_i - \theta)^2$	$\sum_{i=1}^n X_i^2$
Sufficient Statistic	Fisher-Neyman Factorization Theorem
Remark	Sufficient Statistic for Exponential Family
Minimal Sufficient Statistic	Ancillary Statistic
Complete Statistic	Basu's Theorem

$\sum_{i=1}^n X_i^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + \frac{(\sum_{i=1}^n X_i)^2}{n}$	$\sum_{i=1}^n (X_i - \theta)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n (\bar{X} - \theta)^2$
Let $X_1, \dots, X_n$ have joint pdf/pmf $f(\mathbf{x} \theta)$. A statistic $T = T(\mathbf{X})$ is sufficient for θ iff the joint distribution can be factored as $f((\mathbf{x} \theta)) = g(T(\mathbf{x}) \theta) h(\mathbf{x}) \quad \forall \theta \in \Theta$ where g depends on $\mathbf{x}$ only through $T(\mathbf{X})$ and h does not depend on θ	A statistic $T(\mathbf{X})$ is a sufficient statistic for θ if the conditional distribution of the sample $\mathbf{X}$ given the value of $T(\mathbf{X})$ does not depend on θ
Let $X_1, \dots, X_n$ be <i>iid</i> observations from pdf/pmf $f(x \boldsymbol{\theta})$ that belongs to an exponential family given by $f(x \boldsymbol{\theta}) = h(x)c(\boldsymbol{\theta}) \exp \left\{ \sum_{j=1}^k w_j(\boldsymbol{\theta}) t_j(x) \right\}$ where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_d), d \leq k$. Then, $T(\mathbf{X}) = \left(\sum_{j=1}^n t_1(X_j), \dots, \sum_{j=1}^n t_k(X_j) \right)$ is sufficient for $\boldsymbol{\theta}$	A 1-1 function of a sufficient statistic T is also sufficient
A statistic $S(\mathbf{X})$ whose distribution does not depend on the parameter θ is called an ancillary statistic	A statistic $T(\mathbf{X})$ is a minimal sufficient statistic if it is sufficient and a function of any other sufficient statistic. Theorem: Let $f(\mathbf{x} \theta), \theta \in \Theta$ be a pdf/pmf of $\mathbf{X} = (X_1, \dots, X_n)$. Suppose there exists a statistic $T(\mathbf{X})$ such that for any two values, say $\mathbf{x}$ and $\mathbf{y}$, the ratio $\frac{f(\mathbf{x} \theta)}{f(\mathbf{y} \theta)}$ is free of θ iff $T(\mathbf{X}) = T(\mathbf{Y})$. Then $T(\mathbf{X})$ is minimal sufficient statistic for θ.
If $T(\mathbf{X})$ is a complete and minimal sufficient statistic, then $T(\mathbf{X})$ is independent of every ancillary statistic.	Let $f(t \theta)$ be a family of pdfs/pmfs for a stat $T(\mathbf{X})$. The family of probability distributions is called complete if $\mathbb{E}[g(T)] = 0 \quad \forall \theta \implies P_\theta(g(T) = 0) = 1 \quad \forall \theta \in \Theta$ Equivalently, $T(\mathbf{X})$ is called a complete statistic

Complete and Minimal Sufficiency	Complete Statistic in Exponential Family
Method of Moments Estimation (MoM/MME)	Maximum Likelihood Estimation (MLE)
Invariance Property of the MLE	Bayes Estimator
Conjugate Family	Mean Squared Error (MSE)
Unbiased Estimator	Cramer-Rao Lower Bound (CRLB)

Let $X_1, \dots, X_n$ be <i>iid</i> observations from an exponential family with pdf/pmf of the form $f(x \boldsymbol{\theta}) = h(x)c(\boldsymbol{\theta}) \exp \left\{ \sum_{j=1}^k w_j(\boldsymbol{\theta}) t_j(x) \right\} \text{ where } \boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_k).$ Then the statistic $T(\mathbf{X}) = \left(\sum_{j=1}^n t_1(X_j), \dots, \sum_{j=1}^n t_k(X_j) \right)$ is complete if $\{w_1(\boldsymbol{\theta}), \dots, w_k(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$ contains an open set in $\mathbb{R}^k$	If a minimal sufficient statistic exists, then any complete statistic is also a minimal sufficient statistic
The estimate $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ is an MLE estimate if $L(\hat{\theta} \mathbf{X}) = \max_{\theta \in \Theta} L(\theta \mathbf{X})$ • Calculus approaches work sometimes • Check boundaries • The MLE may not be unique • The MLE (if unique) is a function of the sufficient statistic	$\mu_j = \mathbb{E} [X_i^j] \text{ and } m_j = \frac{1}{n} \sum_{i=1}^n X_i^j$ Set $\mu_j = m_j \forall j$ and solve for θ • MoM is usually easy to derive and compute • MoM is usually consistent but not efficient
$\hat{\theta}^{\text{Bayes}} = \mathbb{E} [\theta \mathbf{X}] \text{ where } \pi(\theta \mathbf{x}) = \frac{f(\mathbf{x} \theta)\pi(\theta)}{\int_{\Theta} f(\mathbf{x} \theta)\pi(\theta)} \propto f(\mathbf{x} \theta) \pi(\theta)$	If $\hat{\theta}$ is the MLE of θ, then for any function $\tau(\theta)$, the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$
$\text{MSE} = \text{Var}(W) + (\text{Bias}(W))^2 = \text{Var}(W) + (\mathbb{E}[W] - \theta)^2$	Let $\mathcal{F}$ denote the class of pdfs/pmf's $f(x \theta)$. A class Π of prior distributions is a conjugate family for $\mathcal{F}$ if the posterior distribution $\pi(\theta \mathbf{x})$ is in the class $\Pi \forall f \in \mathcal{F}, \pi(\theta) \in \Pi, x \in \mathcal{X}$
Let $X_1, \dots, X_n$ have joint pdf/pmf $f(\mathbf{x} \theta)$ not necessarily <i>iid</i>. Suppose $\frac{\partial}{\partial \theta} \int \dots \int W(\mathbf{X}) f(\mathbf{x} \theta) d\mathbf{x} = \int \dots \int W(\mathbf{X}) \frac{\partial}{\partial \theta} f(\mathbf{x} \theta) d\mathbf{x} \text{ for all functions } W \text{ where } \text{Var}(W(\mathbf{X})) < \infty.$ Let $\tau(\theta) = \mathbb{E}[W(\mathbf{X})]$ then $\text{Var}(W(\mathbf{X})) \geq (\tau'(\theta))^2 / I(\theta)$ $I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(\mathbf{x} \theta) \right)^2 \right]$	An estimator $W(\mathbf{X})$ is unbiased for θ if $\mathbb{E}[W(\mathbf{X})] = \theta$

Corollary for CRLB	Note on $I(\theta)$ for CRLB
Rao-Blackwell Theorem	Uniqueness of UMVUE
Lehman-Scheffe Theorem	Likelihood Ratio Test (LRT)
Rejection Region	Consequences of a Test
Power Function	Size α test

If $f(x \theta)$ is an exponential family then $I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(\mathbf{x} \theta) \right]$	Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x \theta)$ then $\text{Var}(W(\mathbf{X})) \geq (\tau'(\theta))^2 / n I_1(\theta) \text{ where}$ $I_1(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(x \theta) \right)^2 \right]$
If W is the UMVUE of $\tau(\theta)$, then W is unique	Let W be any unbiased estimator of $\tau(\theta)$ and let T be a sufficient statistic for θ. Define $\phi(T) = \mathbb{E}[W T]$. Then $\mathbb{E}[\phi(T)] = \tau(\theta)$ and $\text{Var}(\phi(T)) \leq \text{Var}(W) \quad \forall \theta \in \Theta$; that is $\phi(T)$ is a uniformly better unbiased estimator of $\tau(\theta)$.
$\lambda(\mathbf{x}) = \frac{\hat{\theta}_{\text{res}}}{\hat{\theta}}$ $\phi(\mathbf{x}) = \begin{cases} 1 & \lambda(\mathbf{x}) \leq c \\ 0 & \lambda(\mathbf{x}) > c \end{cases}$	Let $X_1, \dots, X_n$ have joint distribution $f(\mathbf{x} \theta), \theta \in \Theta$. Suppose that T is a complete sufficient statistic and $\mathbb{E}[W(T)] = \tau(\theta) \quad \forall \theta \in \Theta$. This implies that $W(T)$ is the unique function of T that is an unbiased estimator of $\tau(\theta)$ and hence it is the UMVUE of $\tau(\theta)$.
LOOK AT DECISION MATRIX	$\{\mathbf{x} : \lambda(\mathbf{x}) \leq c\}$
A test with power function $\beta(\theta)$ such that $\sup_{\theta \in \Theta_0} \beta(\theta) = \alpha$	$\beta(\theta) = P_{\theta}(\mathbf{x} \in \mathcal{R}) = \mathbb{E}[\phi(\mathbf{x})]$

Level α test	Unbiased Test
Most Powerful Test (MPT)	Neyman-Pearson (NP) Lemma
Randomized Test	Remark
Remark	UMP level- α test
Monotone Likelihood Ratio (MLR)	Karlin-Rubin (KR) Theorem

A test with power function $\beta(\theta)$ is unbiased if $\beta(\theta') \geq \beta(\theta'') \quad \forall \theta' \in \Theta_1, \theta'' \in \Theta_0 \implies \beta(\theta') \geq \alpha \quad \forall \theta' \in \Theta_1$	A test with power function $\beta(\theta)$ such that $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$
Consider $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ Reject if $f(\mathbf{x} \theta_1) > kf(\mathbf{x} \theta_0)$ This is an MP level-α test	For a simple hypothesis, a test ϕ is MP level α test if $\mathbb{E}[\phi(\mathbf{x})] \leq \alpha \text{ AND } \mathbb{E}_{\theta_1}[\phi(\mathbf{x})] \geq \mathbb{E}_{\theta_1}[\phi^*(\mathbf{x})]$
Neyman-Pearson Lemma does not assume <i>iid</i>, it just needs the joint	Reject H_0 with some probability γ rather than deterministically, so $\phi(\mathbf{x}) = \begin{cases} 1 & \text{Always reject } H_0 \\ 0 & \text{Never reject } H_0 \\ \gamma & \text{Reject } H_0 \text{ with probability } \gamma \end{cases}$ This is mainly used for discrete distributions
A test ϕ is UMP level-α test if $\mathbb{E}_{\theta_0}[\phi(\mathbf{x})] \leq \alpha \quad \forall \theta \in \Theta_0$ $\mathbb{E}_{\theta}[\phi(\mathbf{x})] > \mathbb{E}_{\theta}[\phi^*(\mathbf{x})] \quad \forall \theta \in \Theta_1$ for any other level-α test ϕ^*	In general, the power of the MP Test is greater than α (and is always greater than or equal to α).
Suppose $\{f(\mathbf{x} \theta) \theta \in \Theta\}$ has MLR in $T(\mathbf{X})$. Consider testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$. Then there exists a UMP level-α test of H_0 vs. H_1 given by $\phi(\mathbf{x}) = \begin{cases} 1 & T(\mathbf{X}) > c \\ \gamma & T(\mathbf{X}) = c \\ 0 & T(\mathbf{X}) < c \end{cases}$	The family of distributions $\{f(\mathbf{x} \theta) : \theta \in \Theta\}$ is said to have MLR in $T(\mathbf{X})$ if for $\theta_1 < \theta_2$, the ratio $\frac{f(\mathbf{x} \theta_2)}{f(\mathbf{x} \theta_1)}$ is a nondecreasing function of $T(\mathbf{X})$

Uniformly Most Powerful Test (UMPU)	p -value
Risk function for Bayes test	Coverage Probability
Acceptance Region + Confidence Set	Generalized Method of Moments (MoM)
Pivotal Quantity (Pivot)	Location-Scale Pivots
Pivoting a continuous CDF	Pivoting a discrete CDF

If large value of $W(\mathbf{X})$ give evidence of H_1 $p(\mathbf{x}) = \sup_{\theta \in \Theta_0} P(W(\mathbf{X}) \geq W(\mathbf{x}))$ If small value of $W(\mathbf{X})$ give evidence of H_1 $p(\mathbf{x}) = \sup_{\theta \in \Theta_0} P(W(\mathbf{X}) \leq W(\mathbf{x}))$	• Best unbiased test • the largest power for $\theta \in \Theta_1$ among all unbiased level-α tests • for two-sided tests
$P(L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X}))$	If $\theta \in \Theta_0$: $R(\theta, \delta) = c_I \beta(\theta)$ If $\theta \in \Theta_1$: $R(\theta, \delta) = c_{II} (1 - \beta(\theta))$ $\beta(\theta) = P(\mathbf{X} \in R) = P(\delta(\mathbf{x}) = a_1)$
For non-<i>iid</i> data: Take average of expectations $\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]^j$ and set equal to m_j. The first j is shown below $\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n X_i$	Let $R = \left\{ \mathbf{x} : \left \frac{\bar{x} - \theta_0}{\sigma/\sqrt{n}} \right > z_{\alpha/2} \right\}$ then $A(\theta_0) = \left\{ \mathbf{x} : \left \frac{\bar{x} - \theta_0}{\sigma/\sqrt{n}} \right \leq z_{\alpha/2} \right\}$ and $C(\mathbf{x}) = \left\{ \theta : \bar{x} - z_{\alpha/2} \sigma / \sqrt{n} \leq \theta \leq \bar{x} + z_{\alpha/2} \sigma / \sqrt{n} \right\}$ Then $\mathbf{x} \in A(\theta_0) \iff \theta_0 \in C(\mathbf{x})$
CHECK YOUR NOTES AND KNOW THEM	A random variable $Q(\mathbf{x}, \theta)$ is a pivot if the distribution of $Q(\mathbf{x}, \theta)$ is independent of all parameters.
Let T be a statistic with a discrete CDF $F_T(t \theta)$ and let $\alpha_1 + \alpha_2 = \alpha, 0 < \alpha < 1$ where α_1 and α_2 are fixed. 1. $F_T(t \theta)$: a decreasing function of θ $F_T(t \theta_U(t)) = \alpha_1$ AND $F_T((t-1) \theta_L(t)) = 1 - \alpha_2$ 2. $F_T(t \theta)$: an increasing function of θ $F_T((t \theta_U(t)) = 1 - \alpha_2$ AND $F_T(t \theta_L(t)) = \alpha_1$	Let T be a statistic with a continuous CDF $F_T(t \theta)$ and let $\alpha_1 + \alpha_2 = \alpha, 0 < \alpha < 1$ where α_1 and α_2 are fixed. 1. $F_T(t \theta)$: a decreasing function of θ $F_T(t \theta_U(t)) = \alpha_1$ AND $F_T(t \theta_L(t)) = 1 - \alpha_2$ 2. $F_T(t \theta)$: an increasing function of θ $F_T(t \theta_U(t)) = 1 - \alpha_2$ AND $F_T(t \theta_L(t)) = \alpha_1$

Poisson to χ^2 Theorem	Bayes Credible Interval
Bayes HPD Region	Property of Intervals
UMA Confidence Interval	NOTE:
Unbiased Confidence Set	Asymptotic Properties
Markov's Inequality	Consistent Sequence of Estimators

A $1 - \alpha$ Bayes Interval $(a(\mathbf{x}), b(\mathbf{x}))$ is $\int_{a(\mathbf{x})}^{b(\mathbf{x})} \pi(\theta \mathbf{x}) d\theta = 1 - \alpha$ $\iff P(a(\mathbf{x}) \leq \theta \leq b(\mathbf{x})) = 1 - \alpha$	$X \sim \text{Pois}(\mu) \implies P(X \leq x) =$ $P(\text{Gamma}(x+1, 1) > \mu) = P(G > \mu), G \sim$ $\text{Gamma}(x+1, 1) \implies P(2G > 2\mu) =$ $P(\chi_{2(x+1)}^2 > 2\mu)$
- Expected Length (Size) - Coverage Probability	A level $1 - \alpha$ HPD Bayes region for θ is a subset C of Θ such that $C = \{\theta : \pi(\theta \mathbf{x}) \geq k(\alpha)\}$ where $k(\alpha)$ is the largest constant such that $P(C \mathbf{x}) \geq 1 - \alpha$
- UMP test $\iff$ UMA confidence interval - Just invert a UMP test to get the UMA interval	A $1 - \alpha$ confidence interval that minimizes the probability of false coverage
Describe the limiting behavior of a procedure as $n \rightarrow \infty$	- A $1 - \alpha$ confidence set $C(\mathbf{x})$ is unbiased if $P(\theta' \in C(\mathbf{x})) \leq 1 - \alpha \quad \forall \theta = \theta'$ - Obtained by inverting an unbiased test
A sequence of estimators $W_n = W(X_1, \dots, X_n)$ is a consistent sequence of estimators of θ if $\lim_{n \rightarrow \infty} P(W_n - \theta < \epsilon) = 1 \quad \forall \epsilon > 0 \quad \text{AND} \quad \forall \theta \in \Theta$ This mean that W_n becomes close to θ with probability 1 as $n \rightarrow \infty$	$P(g(\mathbf{x}) \geq r) \leq \frac{\mathbb{E}[g(\mathbf{x})]}{r}$

Weak Law of Large Numbers	Consistency by Limiting Variance and Bias
Consistency of MLE	Limiting Variance
Asymptotic Variance (AV)	Asymptotically Efficient Estimator
First Order Delta Method	Second Order Delta Method
Asymptotic Relative Efficiency (ARE)	Multivariate Delta Method

Suppose W_n is a sequence of estimators of θ satisfying 1. $\lim_{n \rightarrow \infty} \text{Var}(W_n) = 0 \quad \forall \theta \in \Theta$ 2. $\lim_{n \rightarrow \infty} \text{Bias}(W_n) = 0 \quad \forall \theta \in \Theta$ Then W_n is a consistent sequence of estimators of θ	Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_X(x)$ with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$ and $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then $\forall \epsilon > 0$ $\lim_{n \rightarrow \infty} P(\bar{X}_n - \mu < \epsilon) = 1 \quad \text{ie} \quad \bar{X}_n \xrightarrow{P} \mu$ This guarantees that $\bar{X}_n$ is a consistent estimator of μ
For an estimator T_n, if $\lim_{n \rightarrow \infty} \{k_n \text{Var}(T_n)\} = \tau^2 < \infty$ where k_n is a sequence of constants. Then τ^2 is the limiting variance	Under the regularity conditions, • $\hat{\theta}$ is a consistent estimator of θ • $\hat{\tau}(\theta)$ is a consistent estimator of $\tau(\theta)$
A sequence of estimators W_n is asymptotically efficient (AE) for a parameter $\tau(\theta)$ if $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{d} N(0, V(\theta))$ where $V(\theta)$ is the asymptotic variance and is equal to the CRLB	For an estimator T_n, suppose $k_n(T_n - \tau(\theta)) \xrightarrow{d} N(0, \sigma^2)$ Then σ^2 is the asymptotic variance of T_n NOTE: The asymptotic variance requires the limiting distribution of T_n to be normal with mean $\tau(\theta)$
Let Y_n be a sequence of random variables such that $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ For a given function g and a specific value of θ, suppose $g'(\theta) = 0$ and $\exists g''(\theta) \neq 0$. Then $\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} \frac{\sigma^2 g''(\theta)}{2} \chi_1^2$	Use when approximating the distribution of a function of an estimator. Let Y_n be a sequence of random variables such that $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ For a given function g and a specific value of θ, suppose $\exists g'(\theta) \neq 0$. Then $\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 g'(\theta)^2)$
Let $\hat{\theta}_n \in \mathbb{R}^k$ be a vector of estimators such that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N_k(0, \Sigma)$ and $g: \mathbb{R}^k \mapsto \mathbb{R}^m$ be differentiable at θ with Jacobian $\mathbf{J} \in \mathbb{R}^{m \times k}$ then $\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} N_m(0, \mathbf{J}\Sigma\mathbf{J}^\top)$	If two estimators W_n and V_n satisfy $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{d} N(0, \sigma_W^2) \quad \text{AND}$ $\sqrt{n}(V_n - \tau(\theta)) \xrightarrow{d} N(0, \sigma_V^2)$ Then the ARE of V_n wrt W_n is $\text{ARE}(V_n, W_n) = \frac{\sigma_W^2}{\sigma_V^2}$

Nice LRT Asymptotic Theorem	Slutzky's Theorem
Wald Test	Score Test
Wald-type Interval Estimation	Score Interval (Wilk's Interval)

Suppose $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} a$ where a is a constant 1. $Y_n X_n \xrightarrow{d} aX$ 2. $X_n + Y_n \xrightarrow{d} X + a$	Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_X(x \boldsymbol{\theta})$, $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subset \mathbb{R}^k$ with $\boldsymbol{\Theta}_0^m \subset \boldsymbol{\Theta}^k$ where $m < k$. Suppose that the distribution $f_X(x \boldsymbol{\theta})$ satisfies all regularity conditions for asymptotic normality of the MLE. Then $-2 \log \lambda(\mathbf{x}) \xrightarrow{d} \chi_{k-m}^2$
Let $S(\theta) = \frac{\partial}{\partial \theta} \log f(\mathbf{x} \theta)$ be the score statistic Recall that $I(\theta)$ is the Fisher information. Then $S_n = \frac{S(\theta_0) - 0}{\sqrt{I_n(\theta_0)}} \xrightarrow{d} N(0, 1)$	Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_X(x \theta)$ We have $\hat{\theta} : \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \tau^2)$ Then the Wald Stat is • $W_n = \frac{\hat{\theta} - \theta}{\tau/\sqrt{n}} \xrightarrow{d} N(0, 1)$ (τ known) • $W_n = \frac{\hat{\theta} - \theta}{\hat{\tau}/\sqrt{n}} \xrightarrow{d} N(0, 1)$ (τ unknown)
$Q(\mathbf{x} \theta) = \frac{\frac{\partial}{\partial \theta} \log f(\mathbf{x} \theta)}{\sqrt{\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(\mathbf{x} \theta) \right]}} \xrightarrow{d} N(0, 1)$ An approximate $1 - \alpha$ score interval for θ is the interval that satisfies $\{\theta : Q(\mathbf{x} \theta) \leq z_{\alpha/2}\}$	Let T_n and S_n be statistics such that $\frac{T_n - \theta}{S_n} \xrightarrow{d} N(0, 1)$ Then an approximate $1 - \alpha$ Wald confidence interval for θ is $T_n \pm Z_{\alpha/2} S_n$